

Colin Lin

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Experience

Stripe

Toronto, ON

Software Engineer

October 2024 - Present

- Built an internal **Python/Java** platform to host **agentic AI** applications, scaling to **170k+ weekly executions** and automating Stripe merchant risk-review workflows with **\$4.25M+** in annualized business impact
- Contributed core components to an internal **LLM gateway** built on LiteLLM, supporting **3B+ tokens/day** across **60+** internal use cases and integrating GPT-5, Gemini, and Claude models
- Built and scaled an internal **LLM chat UI** by extending LibreChat (**React/Node.js**), reaching **6000+ weekly active users** and adding a **Go** proxy for secure authentication and identity
- Reduced **LLM costs by 80%** by implementing prompt caching and execution controls for internal coding agents, saving **\$800k/year** and improving system reliability
- Pioneered an internal **no-code Agent Runtime**, using a **Python** agentic platform and a no-code UI to enable **400+ agents** with **MCP** tool access for non-engineers
- Resolved a high-severity production incident in **74 minutes** by diagnosing **Redis** rate-limiting failures and re-architecting the deployment to stabilize a platform used by **1000+ engineers**

Konrad Group

Toronto, ON

Software Developer

September 2023 - August 2024

- Developed new frontend features for a major airline's flight booking application using **Vue.js**, impacting over **1 million** users
- Built online contests and promotional pages for a major real estate client using **Next.js** and **Contentful CMS**
- Accelerated internal IT processes by **300%** by automating manual tasks using **Python** and **Selenium**
- Mitigated fraudulent contest entries by implementing browser fingerprinting using **JavaScript**

HSBC

Vancouver, BC

Cloud Full Stack Engineer

December 2021 - December 2022

- Developed cloud-based web apps for HSBC Wealth Compass and US Product Onboarding using **React** and **Redux**
- Engineered RESTful APIs and API Orchestration layer using **Java Spring Boot** and **Express.js** microservices deployed on **AWS**
- Reduced maintenance cost by 40% by migrating legacy JavaScript codebase to **TypeScript**
- Worked closely with a cross time-zone **Agile Scrum** team to support systems critical to Retail Banking across North America
- Mentored 3 new employees and interns on frontend development, API design, and advanced product knowledge

University of Toronto

Toronto, ON

Software Developer

September 2021 - February 2022

- Analysed trends across global news sites by constructing data visualizations using **Python**, **pandas**, and **Matplotlib**
- Improved the efficiency of the Python website crawler by 50% by applying filters and caching data
- Served browser-accessible **Jupyter Notebooks** for researchers using Jupyterhub deployed on the cloud

Cineplex Digital Media

Toronto, ON

Software Developer

September 2020 - April 2021

- Led the development of **web applications** and digital signage for clients such as RBC and Scotiabank
- Adapted **HTML**, **CSS**, and **JavaScript** projects to serve over 30 different countries, currencies, and languages
- Automated manual waiver signing process by building a customer-facing portal using **React**

Education

University of Toronto

Toronto, ON

Honours Bachelor of Science in Computer Science

September 2019 - August 2023

- **GPA:** 3.88/4.0
- **Publications:** A Systematic Literature Mapping of COVID-19 Papers in Computer Science Education
- **TA:** CSCC09 Programming on the Web
- **Relevant Courses:** Software Engineering, Data Structures and Algorithms, Databases, Computer and Network Security

Skills

Programming	Python, Go, TypeScript, JavaScript, Java, HTML, CSS, C, Swift, Kotlin, PHP
Frameworks and Libraries	React, Node.js, Redux, Next.js, NumPy, pandas, Spring Boot, Selenium, Angular, Vue.js
Databases	MongoDB, MySQL, PostgreSQL, Redis, Firestore, Neo4j
Tools	Git, Docker, Linux, AWS, Databricks, Postman, nginx, Firebase, Jira